

Subashji Natarajan

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Skills

Languages: Java, Python, HTML, CSS, SQL

Frameworks: Flask, FastAPI, Tensorflow, Langchain

AI & LLM Tools: Google Gemini

Databases: MySQL

Tools: Git, Github

Work Experience

Tech Mahindra, Chennai

Sep 2024 - Present

Associate Software Engineer

- Built a **GenAI-powered BFSI platform** using Flask and React, improving financial insight delivery speed by **40%** and automating **60%** of customer interactions via chatbot workflows.
- Developed and integrated **OpenCV-based Aadhaar/PAN verification**, reducing manual document checks by **50%** and cutting overall verification time by **35%**.
- Enhanced backend reliability and model accuracy across Agile sprints, boosting system performance and user engagement by **30%**.
- Python, React.js, LLM, Google Gemini, Flask, MySQL, Git, FAISS.

Education

Amrita Vishwa Vidyapeetham, Coimbatore

Nov 2020 – Jul 2024

B.Tech. in ECE

CGPA: 7.07/10

Project Work

- **PDF Chatbot Using Gemini (Jan 2024 – Mar 2024):**
 - Created an AI chatbot using LangChain and Gemini API, powered by spaCy and NLTK for natural language understanding.
 - Utilized Pandas for document parsing, improving query accuracy by 40% and reducing response time by 25%.
 - This chatbot was specifically built to read, analyze, and extract insights from reference research papers required for the Sentinel-2 anomaly detection project, supporting the preparation and publication of our IEEE paper.
- **Anomaly Detection in Sentinel-2 Imagery (Sep 2023 – Mar 2024):**
 - Applied **Autoencoders** and **Attention Models** for crop anomaly detection using TensorFlow.
 - Performed data cleaning, feature engineering, and augmentation on multispectral imagery
 - Achieved **60% dimensionality reduction** with tuned hyperparameters and validation strategies.

Awards and Certificates

- **Deep Learning Models and Introduction of Raspberry Pi:** IIT Palakkad.
- **AI White Belt:** Tech Mahindra
- **Google Generative AI Fundamentals:** Google Cloud
- **Machine Learning with Python:** Coursera
- **Publication:** “*Unsupervised Land Use Land Cover Mapping of Sentinel-2 Data using Convolutional Autoencoders*” (IEEE 2024)
- Delivered an internal presentation on “**Introduction to Generative AI**” for training sessions.
- Led technical events and hackathons with 100+ participants and organized AI & IoT workshops that fostered student-driven innovation.